

Challenge (Habanero)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
 (B) $ax + bx^2 + c$
 (C) $ax^2 - bx + c$
 (D) $ax^2 + b$
 (E) $ax + c$

(E) $y = x^2 + 2x + 3$

Q2. A recipe for making cookies calls for x cups of flour, where x satisfies the equation $x^2 + 4x + 4 = 16$. What is the standard form of this equation?

- (A) $x^2 + 4x - 12 = 0$
 (B) $x^2 - 4x + 4 = 0$
 (C) $x^2 - 4x - 12 = 0$
 (D) $x^2 + 4x + 12 = 0$
 (E) $x^2 - 4x + 12 = 0$

(A) $x =$
 (A) $x =$

(A) $x =$

(A) $x =$

Q3. A construction crew is building a rectangular foundation with a quadratic area function $A(x) = 2x^2 + 5x + 1$. What is the vertex of this parabola?

(A) $(-$

$\frac{rac54}{rac98}, -$

(A) $(-$
 $\frac{rac52}{rac92}, -$

(A) $(-$
 $\frac{rac12}{rac92},$

(A) $($
 $\frac{rac54}{rac98},$

(A) $($
 $\frac{rac12}{rac92},$

Q4. Which quadratic function represents a parabola with its vertex at $(2, 3)$?

- (A) $y = (x - 2)^2 + 3$
 (B) $y = -(x - 2)^2 + 3$
 (C) $y = (x + 2)^2 - 3$
 (D) $y = -(x + 2)^2 - 3$

Q5. A baker is making a cake with a quadratic volume function $V(x) = 3x^2 - 2x - 5$. What is the axis of symmetry of this parabola?

(A) $x = -$

$\frac{rac13}{rac13}$

$\frac{rac13}{rac23}$

$\frac{rac13}{rac16}$

$\frac{rac13}{rac53}$

$\frac{rac13}{rac16}$

$\frac{rac13}{rac53}$

$\frac{rac13}{rac53}$

$\frac{rac13}{rac53}$

Q6. Which of the following is a key feature of the quadratic function $f(x) = x^2 + 2x + 1$?

- (A) It has a maximum value
 (B) It has a minimum value
 (C) It has no x -intercepts
 (D) It has no y -intercept
 (E) It is a cubic function

Q7. A contractor is designing a quadratic roof with a function $h(x) = -x^2 + 5x + 2$. What is the y -intercept of this parabola?

- (A) $(y = 2)$
 (B) $(y = 5)$
 (C) $(y = -2)$
 (D) $(y = 0)$
 (E) $(y = 7)$

Q8. A chef is whipping up a quadratic recipe with x as the number of cups of sugar, where x satisfies $x^2 - 3x - 2 = 0$. What are the solutions to this equation?

- (A) $x = -1, x = 2$
 (B) $x = -2, x = 1$
 (C) $x = 1, x = -2$
 (D) $x = -1, x = -2$
 (E) $x = 2, x = 1$

Open Ended Questions (FRQ)

Q9. Solve the quadratic equation $x^2 + 4x + 4 = 0$ and write the solutions in the form $x = a \pm b\sqrt{c}$. Show your work and provide the final answer.

Q10. A baker is creating a cake with a height given by the quadratic function $h(x) = -x^2 + 6x + 5$, where x is the number of hours the cake is baked. Find the vertex of the parabola, the axis of symmetry, and the maximum height of the cake. Provide a diagram and show your work.

Chef's Tips

- Provide a calculator for the students
- Allow students to use a formula sheet
- Encourage students to show their work and provide explanations